Combination of Spectral Angle Cosine and Band Depth Can Reduce the Specular Reflection Effects in Leaf Biochemical Parameters Estimation From Multiangular Hyperspectral Data

Xiao Li, Zhongqiu Sun, Shan Lu, and Kenji Omasa

***Abstract*— Leaf chlorophyll content (LCC) and equivalent water thickness (EWT) are critical indicators of plant physiological state and photosynthetic capacity. Hyperspectral measurements provide technical support for the efficient estimation of these leaf biochemical parameters. However, most current studies ignored the influence of leaf surface specular reflection, which reduces estimation accuracy, in the multiangular (MA) measurements. This study proposed the spectral angle cosine-band depth indices ($SACI_{BD}$), a method that transforms additive specular reflection into multiplicative differences using band depth (BD) and eliminates them via spectral angle cosine (SAC). Based on an MA calibration dataset, the optimal spectral intervals (OSIs) for $SACI_{BD}$ were identified. Then, $SACI_{BD}$ was validated across five independent leaf datasets and canopy imaging datasets obtained through close-range camera and LESS simulated scene. Results demonstrated that $SACI_{BD}$ efficiently captured biochemical-sensitive spectral absorption shape features (OSI_{LCC}: 658–748 nm and OSI_{EWT}: 1872–2026 nm) and eliminated the specular reflection influence on biochemical estimation. Strong linear relationships with both LCC ($R^2 = 0.9$) and EWT ($R^2 = 0.94$), as well as validation results of five leaf datasets (LCC: RMSE = 5.98 $\mu g/cm^2$, $n = 2143$; EWT: RMSE = 0.0031 g/cm^2, $n = 1919$), confirmed the applicability of $SACI_{BD}$ for various measurement strategies and plant species in the estimation of leaf biochemical parameters. When the study was extended to the canopy level, $SACI_{BD}$ achieved similarly good accuracy (LCC: RMSE = 6.98 $\mu g/cm^2$, $n = 104$; EWT: RMSE = 0.0044 g/cm^2, $n = 34$). Thus, the novel combination of SAC and BD provides an effective tool for quantitative plant biochemistry research.**

***Index Terms*— Equivalent water thickness (EWT), leaf chlorophyll content (LCC), multiangular (MA) hyperspectral band depth (BD), spectral angel cosine (SAC), specular reflection.**

I. Introduction

CHLOROPHYLL and water in plant leaves are crucial factors in many biogeochemical processes, such as evapotranspiration, photosynthesis, and net primary productivity [1], [2], [3]. The spatiotemporal variations of leaf chlorophyll content (LCC) and equivalent water thickness (EWT) indicate the physiological state of the plant, including water or nutrient deficiency, disease, and senescence, during the whole growth process [4], [5]. Thus, monitoring these biochemical parameters for ecological, agricultural, and forestry applications is necessary. Traditional laboratory biochemical analysis, however, is sample destructive and time-consuming and cannot be applied to long-term and large-area plant monitoring [6]. Since the measured spectra respond to the changes in leaf biochemical parameters [7], [8], remote sensing is an alternative approach that enables the monitor of plant chlorophyll and water conditions rapidly, continuously, and nondestructively [9], [10]. Hyperspectral measurements contain abundant band information, therefore allowing the accurate estimation of LCC and EWT at the leaf, plant, or stand levels [11], [12], [13] and ultimately serve the research of land surface ecosystem [14].

Physical and empirical methods based on hyperspectral measurements have been used to retrieve or estimate LCC and EWT in many studies [15], [16], [17]. The radiative transfer model, such as PROSPECT, has a precise physical mechanism and needs a variety of physical and chemical parameters as inputs, thus enabling researchers to obtain acceptable inversion accuracy of leaf biochemical parameters [18], [19]. On the contrary, spectral indices are carefully formulated by spectral reflectance at two or more chlorophyll- and water-sensitive or insensitive wavelengths and thus can estimate LCC and EWT easily and simply [20], [21], [22]. In addition, machine learning algorithms (random forest, support vector machine, genetic algorithm, etc.) analyze spectral data with complex algorithms and make high-precision decisions on plant biochemical parameters, such as chlorophyll or water, after learning a large number of training samples [23], [24], [25].

Received 24 October 2024; revised 1 April 2025; accepted 10 June 2025. Date of publication 16 June 2025; date of current version 27 June 2025. This work was supported in part by Jilin Provincial Science and Technology Development Project under Grant 20240402034GH and Grant 20240101047JC, in part by the National Natural Science Foundation of China under Grant 42471377, in part by China Postdoctoral Science Foundation under Grant 2024M760406, and in part by the Fundamental Research Funds for the Central Universities under Grant 2412025QD024. *(Corresponding authors: Zhongqiu Sun; Shan Lu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3580105

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Most of the previous studies estimated chlorophyll or water content based on the assumption that the internal constituents of the leaf determine the spectral characteristics. However, the spectral characteristics of a single leaf, as the data basis of these methods, are dominated by both leaf internal and surface structural properties [26], and the latter reflects incident light in different directions in hemispherical space with different magnitude, and thus the detected directional reflectance factors vary greatly with measurement geometries [27], [28]. Therefore, understanding how the lights scattered from the leaf interior and reflected from the leaf surface affects the estimation of leaf biochemical parameters is of great value for plant remote sensing applications [26], [29].

Studies have shown that the light reflected by leaf (i.e., total spectral intensity) consists of diffuse and specular parts, which jointly affect the spectral characteristics detected by sensors [30], [31], [32]. The diffuse reflection generated from the bulk of leaf tissue is mainly affected by internal biochemical parameters and, therefore, isotropic [26]. For example, pigments, especially chlorophyll, are the main factors to affect the spectral absorption at 675 nm in visible wavelength [33], [34], while water primarily affects the reflection at 970, 1200, 1450, and 1950 nm in shortwave-infrared wavelength [35], [36]. The specular reflection does not interact with internal biochemical parameters and occurs only on the leaf surface [29], which is independent of wavelength and irrelative to chlorophyll and water. Moreover, there is an inherent angle effect of specular reflection, which contributes to the distribution and magnitude of the directional reflectance factor [37]; that is, the anisotropic spectral intensity shows additive differences among different incident-observation geometries [38], [39], [40]. Such spectral intensity differences caused by additive specular variables at various viewing zenith angles (VZAs) may reduce the inversion or estimation accuracy of LCC and EWT [41], [42].

To mitigate the influence of specular reflection at different VZAs on leaf biochemical estimation, researchers have optimized the physical and empirical models: PROCOSINE model, as an example, considered the specular reflection by introducing a wavelength-independent constant specular parameter [43]. PROCWT [44] and PROSDM [42] were also proposed to remove the specular reflection and to improve the biochemical parameter estimation. However, the reflectance factors only included weak specular reflection because they were measured by the leaf clip equipped with the spectrometer, which obtains the bidirectional reflectance factor (BRF) from the backward scattering directions; thus, they have small difference from directional hemisphere reflectance factor (DHRF) [26], [29], [45]. In fact, the effectiveness of specular reflection removal needs to be validated in a dataset including various intensities of BRF at different VZAs. Additionally, spectral indices are more convenient and widely used to estimate LCC and EWT compared with the physical models that strictly limit the inversion process as well as the machine learning algorithms that train a significant number of samples leading to high time and space complexity [46], [47]. For example, the proposed difference ratio indices removed specular reflection by the subtraction of several specific wavelengths (i.e., eliminating the additive differences of spectral intensity) [34], [40], [41], [48], [49]. However, these spectral indices only used a small number of wavelengths, and the majority of the spectrum was discarded [47], [50].

To take full advantage of the spectral information of spectroscopic datasets, shape indices were developed to extract shape-related information from contiguous spectral signatures for a specific spectral region, that is then correlated with a biophysical or biochemical variable [17]. Spectral angle cosine (SAC) is one of them that may solve the problem of inaccurately estimating leaf biochemical parameters by eliminating uncertainty caused by the reflectance intensity [51]. The principle of SAC is to calculate the cosine of the angle between two "spectral vectors" in n-dimensional space to quantify their spectral shape similarity [52], [53]. Studies have shown that SAC is not affected by the multiplicative change of spectral intensity (corresponding to the linear scaling of a spectral vector) caused by solar illumination, terrain shadow, or atmosphere, which is usually applied to image classification and achieves relatively high accuracy [54], [55], [56], [57]. Researchers have tried to combine leaf hyperspectral reflectance with the SAC method to estimate EWT from different plant species [58]. Although high accuracy was found [58], the calibration and validation of the proposed models were just based on the leaf spectral reflection taken from the integrating sphere with little specular reflectance. In high spatial resolution imaging detections (such as uncrewed aerial vehicles carrying hyperspectral image sensors with centimeter-level spatial resolution), the wavelength-independent specular components may occupy and lead to additive differences in reflection intensity of a single leaf due to the randomly distributed leaf orientations and various incident-observation geometries caused by sun positions and sensors' field of view [39]; thus, the specular reflection reduce the estimation accuracy of leaf biochemical parameters [16], [59], [60]. In the case that the additive specular reflections in the multiangular (MA) measurements can be transformed into multiplicative variables, the SAC method may estimate LCC and EWT with good accuracy, then the application possibility for high-precision estimation of leaf biochemical parameters under arbitrary VZAs will be provided.

Band depth (BD) is a preprocessed spectrum that quantifies the relative depth of absorption features by normalizing the original spectral reflection to its continuum, which is typically derived through linear interpolation or curve fitting across the absorption feature [61]. BD highlights the plant absorption characteristics while mitigating nonspecific spectral variations [61], [62]. In MA measurement scenarios, where additive specular reflections are inevitable, the MA BD is calculated based on the ratio of MA spectral reflection to the corresponding continuum. Since both the original spectral reflections and their continuum contain additive specular reflections, this ratio may transform these additive specular reflections into multiplicative variables, making the MA BD consistent with the theory of SAC.

Therefore, we first converted the MA reflection with additive characterization into the MA BD with multiplicative characterization. Subsequently, the SAC method and MA BD

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were combined to eliminate the specular reflection influence. The SAC-BD indices ($SACI_{BD}$) within optimal spectral intervals (OSIs), which is strongly related to LCC and EWT, were proposed and compared to some existing spectral indices based on an MA measurement dataset. Finally, the effectiveness and transferability of our method were further validated by five independent leaf spectral datasets acquired in other regions of the globe, and the two plant canopy images acquired by close-range hyperspectral camera and LESS simulation scene [63].

II. Materials and Methods

A. SAC and MA BD

Leaf BRF across all directions in hemispherical space is controlled by both leaf internal biochemical parameters and surface structural properties [26]. On the one hand, the specific absorption coefficient and contents of leaf biochemical parameters (LCC, EWT, leaf mass per area, etc.) primarily affect diffuse reflection from the internal leaf, which shapes the spectral characteristics of BRF [64], [65]. On the other hand, species-dependent surface structural properties govern specular reflection from the leaf surface, which results in anisotropic variation in BRF [27], [30]. The simulated BRF from the BRF model, PROSPECULAR [66], indicates that anisotropic variations due to specular reflections exhibit "additive" differences that are independent of biochemical parameters and wavelength [see Fig. 1(a)]. To enhance the accuracy of leaf biochemical parameter estimation, these additive differences should be removed [67].

SAC [51] quantifies the shape similarity between two spectra by calculating the cosine of the angle ($\cos\alpha$) between the target (T) and reference spectral (Ref) vectors as follows:

$$\cos\alpha = \frac{\sum_{i=1}^{i=n} \mathrm{Ref}_i T_i}{\sqrt{\sum_{i=1}^{i=n} (\mathrm{Ref}_i)^2 \sum_{i=1}^{i=n} (T_i)^2}} \tag{1}$$

where n is the number of spectral bands, and i corresponds to the individual spectral band ranging from 1 to the number of bands n.

In previous studies, SAC was usually used for feature classification of remote sensing images [53], [55], [57]. The smaller the α, the larger the $\cos\alpha$, the better the similarity between two spectral shapes, and the higher likelihood of them belonging to the same target.

Since SAC is derived from the vector product formula [54], when the target spectral vector scales linearly ($T_{1^*} = 2^*T_0$), i.e., in the case of a multiplicative difference, the $\cos\alpha$ remains unchanged [54] [see Fig. 1(e) and (f)]. In this case, SAC can eliminate the differences in spectral intensity caused by external factors such as topography and shadow on the ground, then achieve relatively high classification accuracy [52], [68]. However, SAC cannot address the additive differences caused by specular reflection among different VZAs [see Fig. 1(c) and (d)].

BD, as an indicator of leaf absorption characteristics, has been applied in plant spectroscopy [61], [62]. BD [see Fig. 1(b)] is calculated by subtracting the ratio of BRF to the continuum [C, dashed lines in Fig. 1(a)] from 1

$$\begin{aligned} \mathrm{BD} &= 1 - \mathrm{CR} \\ &= 1 - \frac{\mathrm{BRF}}{C} \end{aligned} \tag{2}$$

where CR represents continuum removal.

When BD is applied to BRF with additive specular reflection [$\mathrm{BRF}_{\mathrm{additive}}$, (3)], $\mathrm{BD}_{\mathrm{additive}}$ can be calculated as follows:

$$\begin{aligned} \mathrm{BD}_{\mathrm{additive}} &= 1 - \frac{\mathrm{BRF}_{\mathrm{additive}}}{C_{\mathrm{additive}}} \\ &= 1 - \frac{\mathrm{BRF} + \mathrm{constant}}{C + \mathrm{constant}} \end{aligned} \tag{3}$$

where $\mathrm{BRF}_{\mathrm{additive}}$ and $\mathrm{C}_{\mathrm{additive}}$ represent the BRF and C influenced by additive specular reflection, respectively.

The $\mathrm{BD}_{\mathrm{additive}}$ can be related to BD by a multiplicative factor associated with the additive specular reflection (constant) by dividing (3) by (2), as shown as follows:

$$\begin{aligned} \frac{\mathrm{BD}_{\mathrm{additive}}}{\mathrm{BD}} &= \frac{1 - \frac{\mathrm{BRF}+\mathrm{constant}}{C+\mathrm{constant}}}{1 - \frac{\mathrm{BRF}}{C}} \\ &= \frac{C - \mathrm{BRF}}{C + \mathrm{constant}} \times \frac{C}{C - \mathrm{BRF}} \\ &= \frac{C}{C + \mathrm{constant}} \end{aligned} \tag{4}$$

$$\mathrm{BD}_{\mathrm{additive}} = \mathrm{BD}^* \frac{C}{C + \mathrm{constant}}. \tag{5}$$

By applying BD [see (2)–(3)], the additive specular difference in $\mathrm{BRF}_{\mathrm{additive}}$ ($\mathrm{BRF}_{\mathrm{additive}} = \mathrm{BRF} + \mathrm{constant}$) can be transformed into a multiplicate difference in $\mathrm{BD}_{\mathrm{additive}}$ [as shown in the second term of (5)]. The specular reflections in MA BD within some specific wavelength regions are now considered as the multiplicative difference [see Fig. 1(b)], which falls in with the SAC method's theoretical scheme [see Fig. 1(e) and (f)]. Therefore, combining BD with SAC may effectively mitigate the influence of specular reflection when estimating biochemical parameters.

When applying the combination of SAC and MA BD ($\mathrm{SAC}_{\mathrm{BD}}$) to relate to LCC and EWT, it is necessary to determine the reference spectrum. Chlorophyll and water absorption curve (k_{chl} and k_w) characterize the absorption capacity in the ranges of 400–780 and 800–2450 nm, respectively [69], [70], [71], [72]. The in vivo absorption curves (k_{chl} and k_w), simulated by the PROSPECT-5 model, were used as the reference spectra to ensure consistency with leaf reflection measurements, as absorption curves measured in solvents exhibit spectral shifts (~10 nm) compared to in vivo conditions [73], [74]. To align their dimensionality with BD [75], these absorption curves simulated by PROSPECT-5 were normalized to the range 0–1 (Nk_{chl} and Nk_w). These normalized absorption curves served as the reference spectra to quantify the spectral "shape mode" variations of BD caused by biochemical parameters content

$$Nk_{\mathrm{chl}} = [k_{\mathrm{chl}} - \min(k_{\mathrm{chl}})]/[\max(k_{\mathrm{chl}}) - \min(k_{\mathrm{chl}})] \tag{6}$$

$$Nk_w = [k_w - \min(k_w)]/[\max(k_w) - \min(k_w)] \tag{7}$$

where $\min(k_{\mathrm{chl}})$ and $\max(k_{\mathrm{chl}})$ are the minima and maximum values of k_{chl}; $\min(k_w)$ and $\max(k_w)$ are the minima and

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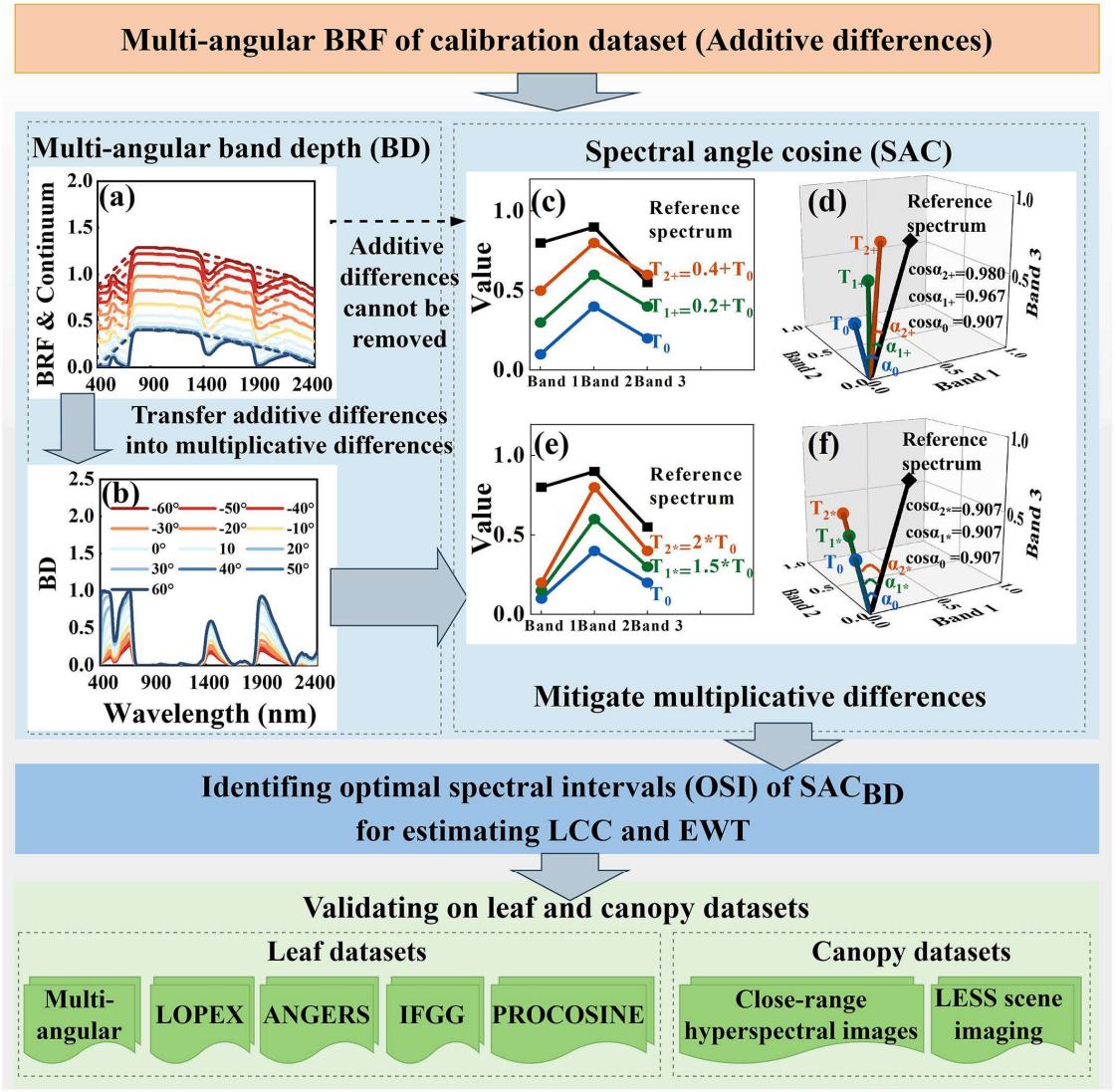


Fig. 1. Schematics for estimating leaf traits (LCC and EWT) using the combination of SAC and MA BD. Because SAC was influenced by additive specular differences (c) and (d), the MA BRF [(a), −60° to 60° VZAs, solid lines] with additive difference (specular reflection) was first transformed into MA BD (b) with multiplicative difference through continuum removal [(a), dashed lines]. SAC, which calculates the cosine of angle between reference vector and target vectors [denoted by T_{j*}, subscript j represent target in (e) and (f)], was applied to the MA BD (e) and (f) within OSIs to eliminate multiplicative differences. The SAC_{BD} within the OSI was then validated across five leaf datasets and two canopy datasets.

maximum values of k_w. By using leaf MA BD as "target spectra" and Nk_{chl} or Nk_w as "reference spectrum," their spectral shape similarity, quantified by SAC within specific spectral intervals, can be used to estimate LCC or EWT under different measurement conditions.

The $SACI_{BD}$ are defined based on the dot product formula as follows:

$$SACI_{BD} = \cos[\alpha(Nk_{SI}, BD_{SI})] = \frac{Nk_{SI} \cdot BD_{SI}}{(\|Nk_{SI}\|) \cdot (\|BD_{SI}\|)} \quad (8)$$

where subscript SI denotes the spectral interval, including start and end wavelength positions. Nk_{SI} and BD_{SI} represent the reference spectrum vectors and target BD spectrum vectors within SI, respectively. $SACI_{BD}$ expresses the cosine of the angle (α) between the reference spectrum and target BD spectrum. To best relate $SACI_{BD}$ to LCC and EWT, we determined the OSIs from all possible SIs. The OSI for LCC (OSI_{LCC}) was selected by iterating through 72 390 combinations (400–780 nm in steps of 1 nm, 381 × 380/2). The OSI for EWT (OSI_{EWT}), were determined by iterating through 1 362 075 combinations (800–2450 nm in steps of 1 nm, 1651 × 1650/2). The spectral intervals with the highest determination coefficient (R^2) from all the iterations were selected as optimal intervals (OSI_{LCC} and OSI_{EWT}).

$SACI_{BD}$ (OSI_{LCC}) and $SACI_{BD}$ (OSI_{EWT}) are expected to achieve the best relationships with LCC and EWT while minimizing the influences of specular reflections from leaf surfaces at various VZAs. The empirical models for estimating these two biochemical parameters were derived from the laboratory MA BD calibration dataset (see Section II-B). Subsequently, these models were validated using five leaf datasets (see Section II-C) and two canopy datasets (obtained through close-range camera and LESS simulated scene, Section II-D). Model performance was evaluated by root mean square error (RMSE), ($\mu g/m^2$ for LCC, and g/m^2 for EWT) between estimated and measured biochemical parameters.

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TABLE I
EXISTING SPECTRAL INDICES EVALUATED IN THIS STUDY. R_Λ REPRESENTS BRF AT THE WAVELENGTH Λ

Biochemical	Type	Index	Reference
LCC	SR	R_{750}/R_{550}	[76]
		R_{860}/R_{550}	[77]
		R_{750}/R_{710}	[78]
	ND	$(R_{750}–R_{705})/(R_{750}+R_{705})$	[79]
		$(R_{800}–R_{650})/(R_{800}+R_{650})$	[79]
		$(R_{800}–R_{635})/(R_{800}+R_{635})$	[80]
	D or DR	D_{730}	[81]
		$(R_{850}–R_{710})/(R_{850}–R_{680})$	[48]
		$(R_{750}–R_{445})/(R_{705}–R_{445})$	[82]
		$(R_{720}–R_{761})/(R_{720}–R_{672})$	[41]
	others	$(R_{750}–R_{705})/(R_{750}+R_{705}–$	[82]
		$R_{705}/(R_{717}+R_{491})$	[82]
		$1/R_{700}$	[83]
		$1/R_{550}–1/R_{750}$	[83]
		$1/R_{700}–1/R_{750}$	[83]
EWT	SR	R_{1300}/R_{1450}	[84]
		R_{900}/R_{970}	[85]
		R_{860}/R_{1240}	[86]
		R_{1670}/R_{1880}	[87]
		R_{860}/R_{1650}	[88]
	ND	$(R_{860}–R_{1240})/(R_{860}+R_{1240})$	[89]
		$(R_{820}–R_{1600})/(R_{820}+R_{1600})$	[90]
		$(R_{860}–R_{1640})/(R_{860}+R_{1640})$	[91]
		$(R_{1062}–R_{1393})/(R_{1062}+R_{1393})$	[92]
	D or DR	$R_{1065}–R_{1310}$	[93]
		$(R_{850}–R_{2218})/(R_{850}–R_{1928})$	[48]
		$(R_{850}–R_{1788})/(R_{850}–R_{1928})$	[48]
		$(R_{1271}–R_{410})/(R_{1342}–R_{410})$	[49]
	others	$2\times R_{1530}–R_{1005}–R_{2055}$	[93]
		$2\times R_{1405}–R_{925}–R_{1885}$	[87]

For comparison, the SAC combined with BRF ($SACI_{BRF}$) was also evaluated for estimating biochemical parameters. Additionally, several existing spectral indices, including simple ratio (SR), normalized difference (ND), derivative (D) or difference ratio (DR), and others, were employed to estimate LCC and EWT (see Table I). It should be noted that R_λ was used as a simplified representation of BRF for spectral indices in this context.

B. Measurement Dataset

To calibrate the $SACI_{BD}$ models and compare them with $SACI_{BRF}$ and existing spectral indices, an MA dataset was measured in laboratory.

1) Measurement of Leaf MA Hyperspectral Reflection: A laboratory dataset of leaves (465 leaf samples in total) was collected at Northeast Normal University in Changchun, Jilin Province, China, during the whole plant growing season (from young, mature, to senescent) in 2018–2020. Leaves from 12 plant species (nine trees, two shrubs, and one liana) possessed diverse leaf surface structural properties and a wide range of biochemical content, which may generate variable bidirectional reflection characteristics under different incident-observation geometries. Due to minor operational errors during LCC experimental procedures, such as grinding and extraction, some LCC data were lost or could not be reliably obtained, resulting in differences in sample numbers between EWT and LCC. The dataset was randomly separated into calibration and validation datasets, as detailed in Table II. For EWT, 317 samples from the dataset were used to establish the model while the remaining 148 samples were used for model validation; and for LCC, 297 samples for model calibration and 145 samples for validation. The calibration dataset was used to establish the linear relationship between $SACI_{BD}$ and LCC/EWT, while the validation dataset, along with four additional datasets (see Section II-C), was used to assess the performance of $SACI_{BD}$.

The collected leaf samples were properly stored (sealed and refrigerated) and quickly transferred into the laboratory for MA hyperspectral measurements. The Northeast Normal University Laboratory Goniospectrometer System (NENULGS) [94], equipped with an artificial light source, an ASD spectroradiometer (Analytical Spectral Devices FieldSpec 4, Boulder, CO, USA), and a goniometer, was used to obtain MA hyperspectral BRF in the range of 350–2500 nm. BRF is defined as the ratio of radiance reflected from the leaf (dL_{sample}) to the radiance reflected from the Spectralon reference ($dL_{reference}$) under the same source-viewing geometries [95]

$$\mathrm{BRF}(\lambda,\theta_s,\varphi_s,\theta_v,\varphi_v)=\frac{dL_{\mathrm{sample}}(\lambda,\theta_s,\varphi_s,\theta_v,\varphi_v)}{dL_{\mathrm{reference}}(\lambda,\theta_s,\varphi_s,\theta_v,\varphi_v)}\rho_\lambda \quad (9)$$

where λ is the wavelength, θ_s is the source zenith angle (SZA), θ_v is the VZA, ϕ_s is the incident azimuth angle, ϕ_v is the viewing azimuth angle, and ρ_λ is ranged from 0.9284 to 0.9926 provided by the manufacturer of the Spectralon.

To explain the effect of measurement geometries on leaf spectra and further on the estimation of biochemical parameters, SZA was set at 30°, 40°, or 50° (corresponding to local solar zenith angles in different seasons). Under each SZA, the VZAs were set as −60° ∼60° with a 10° interval in the principal plane ("−" represents forward scattering directions and "0°" represents nadir direction and others represent backward scattering directions) (see Table II). Note that because of the structural limitation of NENULGS (8° is the smallest phase angle), VZA cannot be equal to SZA in the backward scattering direction; thus, each leaf sample had hyperspectral data from 12 VZAs.

During the measurement, the leaf sample was placed in the center of a black object stage to eliminate the background influence, and the vein was parallel to the principal plane. The ASD spectrometer had a 6° field of view and was kept 20 cm away from the leaf sample; thus, the detection footprint size inside the leaf area varied with different VZAs, from a circle with a diameter of 2.1 cm (0° VZA) to an ellipse with a long axis of 4.2 cm (−60° or 60° VZA). The time of MA hyperspectral measurement for a leaf sample did not exceed 20 min, and we assumed no significant changes in leaf biochemical properties, and BRF measurement was not affected.

The spectral noises of the measured BRF were further reduced: 1) using the Viewspec software provided by the ASD manufacturer to remove the discontinuities at 1000 and 1800 nm caused by the design limitation of the

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TABLE II

STATISTICAL OF THE LABORATORY CALIBRATION AND VALIDATION DATASET, INCLUDING PLANT SPECIES NAMES (WITH ABBREVIATIONS), THE NUMBER OF LEAF SAMPLES, MEASUREMENT GEOMETRIES, AND STATISTICS OF LCC AND EWT. CALIBRATION AND VALIDATION SAMPLES WERE USED TO ESTABLISH AND VALIDATE THE RELATIONSHIPS BETWEEN $SACI_{BD}$ WITHIN OSI AND BIOCHEMICAL PARAMETERS, RESPECTIVELY

Plant species	Number of samples for EWT (LCC)	SZA (°)	VZA (°) (10° interval)	LCC (μg/cm²)			EWT (g/cm²)		
				Mean	Min	Max	Mean	Min	Max
Calibration dataset									
Citrus limon (L.) Burm. F. (*Cb*)	18 (18)	30	-60 ~ 20, 40 ~ 60	62.08	20.22	99.03	0.0149	0.0095	0.0189
Schefflera microphylla Merr. (*Sm*)	88 (75)	30, 40, 50	-60 ~ 60	41.18	0.36	79.7	0.0327	0.0162	0.0434
Pachira aquatica (*Pa*)	52 (52)			24.31	0.9	48.15	0.0108	0.0026	0.0157
Populus L. (*Pl*)	30 (30)			34.47	0.95	86.39	0.0104	0.0065	0.015
Acer saccharum Marsh (*Am*)	31 (31)			35.64	0.18	73.05	0.0087	0.0055	0.0116
Juglans (*Ju*)	13 (13)			24.47	0.36	60.41	0.0089	0.0071	0.0104
Armeniaca vulgaris Lam. (*Al*)	22 (22)	40	-60 ~ 30, 50 ~ 60	45.65	0.59	87.8	0.0106	0.0078	0.015
Swida alba Opiz (*Sw*)	20 (20)			27.8	0.18	68.6	0.009	0.0068	0.0122
Prunus padus L. (*Pr*)	23 (21)			32.09	1.36	72.33	0.0078	0.006	0.0098
Epipremnum aureum (*Ea*)	20 (15)			27.27	2.08	56.53	0.0253	0.0184	0.0294
Total	317 (297)	30, 40, 50	-60 ~ 60	35.61	0.18	99.03	0.0174	0.0026	0.0434
Validation dataset									
Acer saccharum Marsh *(Am)*	25 (24)			37.05	0.18	87.85	0.0083	0.0035	0.0116
Juglans (Ju)	56 (56)	40, 50	-60 ~ 60	28.63	3.62	64.26	0.0087	0.0049	0.0105
Epipremnum aureum (Ea)	11 (11)			30.42	4.16	51.65	0.0176	0.0026	0.0279
Populus L. *(Pl)*	8 (8)			66.74	40.24	92.73	0.0111	0.0094	0.0127
Schefflera microphylla Merr. *(Sm)*	16 (16)			34.74	10.27	61.07	0.0308	0.0273	0.0345
Pachira aquatica (Pa)	4 (4)	40	-60 ~ 30, 50 ~ 60	17.3	5.11	17.3	0.0113	0.0097	0.013
Morus alba L. *(Ml)*	2 (2)			18.13	5.52	30.73	0.0085	0.0075	0.0094
Syzygium aromaticum (Sa)	9 (9)			19.96	4.34	36.97	0.0147	0.0116	0.0182
Swida alba Opiz *(Sw)*	17 (15)			34.19	3.03	71.15	0.0085	0.0056	0.0112
Total	148 (145)	40, 50	-60 ~ 60	32.52	0.18	92.73	0.0121	0.0026	0.0345

TABLE III

FOUR MEASURED DATASETS AND A PROCOSINE MODEL SIMULATED DATASET WERE USED TO VALIDATE THE EMPIRICAL MODELS BETWEEN $SACI_{BD}$ AND BIOCHEMICAL PARAMETERS

Dataset	Spectrophotometer	Number of Samples for EWT (LCC)	LCC (μg/cm²)			EWT (g/cm²)			Reference
			Mean	Min	Max	Mean	Min	Max	
MA	ASD FieldSpec4	148 (145)	32.52	0.18	92.73	0.0121	0.0026	0.0345	-
LOPEX	Perkin Elmer Lambda 19	325 (-)	-	-	-	0.0113	0.0010	0.0525	[103]
ANGERS	ASD FieldSpec	276 (276)	33.88	0.78	106.72	0.0116	0.0044	0.0340	[77]
IFGG	ASD FieldSpec3	630 (734)	32.54	8.14	70.43	0.0157	0.0045	0.0580	[104]
PROCOSINE	-	764 (764)	35.06	0.90	102.28	0.0132	0.0023	0.0492	[43]

instrument [96], [97] and 2) using the Savitzky–Golay polynomial least-square function to smooth the spectra in the range of 400–2450 nm (the spectral range used in this study) with a second-order and 20-nm bandwidth window [98].

2) LCC and EWT Measurement: Biochemical measurements were performed immediately in a dark room after spectral measurements. Each leaf sample was divided into two parts: 1) cutting three circular pieces with a diameter of 6 mm in the spectral measurement area to measure LCC and 2) using the remaining leaf to measure EWT.

For LCC measurements, the circular parts were thoroughly ground, and 95% ethanol was used to extract leaf pigment [99]. Then, the absorption of supernatant in the mixed solution was measured using Lambda 900 spectrophotometer (Perkin–Elmer, Waltham, MA, USA), and LCC (μg/cm^2) was determined based on the methods described by Wintermans and De Mots [100].

For EWT measurements, the remaining part provided the data of leaf fresh weight (FW, g) (measured immediately after cutting), leaf area (A, cm^2), and leaf dry weight (DW, g) (dried at 80 °C for 36 h). EWT is defined as the ratio of water quantity to leaf area

$$\mathrm{EWT} = (\mathrm{FW} - \mathrm{DW})/A. \tag{10}$$

The laboratory datasets used for model calibration and validation contain a wide range of LCC and EWT, and the statistics are shown in Table II.

C. Leaf Spectral Datasets Used for Validation

To validate $SACI_{BD}$ models calibrated using the MA dataset at the leaf level, five datasets (see Table III) were included: an MA validation dataset separated from the measurement dataset (see Section II-B), three public datasets, and a PROCOSINE model simulated dataset.

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The first is our laboratory MA dataset. As described in Table II (labeled as "validation dataset") in Section II-B, 148 leaf samples were used for validating the EWT model and 145 for the LCC model. This dataset tested the effectiveness of the regression models in different VZAs.

The second is the leaf optical properties experiment (LOPEX) dataset measured by the Joint Research Center of the European Commission in 1993, including 330 leaf samples of 45 species [101]. This dataset was not used for LCC validation due to the defects of the chlorophyll measurement strategy, while 325 samples were used to validate the empirical model of EWT. The third is the ANGERS dataset measured in Angers, France, in 2003, including 276 leaf samples from 43 plant species [75]. These two datasets measured the DHRF of leaves in the laboratory by the spectrophotometer equipped with an integrating sphere.

The fourth is the IFGG dataset collected by the Institute of Geography and Geoecology (IFGG) of Karlsruhe Institute of Technology in Germany in 2016 and 2017 [102], which has 734 leaf samples with LCC data but 630 with EWT. The ASD spectrometer with a leaf clip was used to measure the BRF of leaves. LOPEX, ANGERS, and IFGG datasets (see Table III) have been used for many spectral-biochemical quantitative studies of plant leaves [103], [104], [105].

The fifth is a leaf BRF dataset simulated by the PROCOSINE model [43]. PROCOSINE is designed to simulate the [$\mathrm{BRF}_{\mathrm{PROCOSINE}} = (\cos\theta_i/\cos\theta_s) * (\mathrm{DHRF}_{\mathrm{PROSPECT-5}} + b)$] of leaves by integrating the PROSPECT-5 model, which simulates the directional-hemispherical reflectance factor (DHRF), with the COSINE model. The term b represents a wavelength-independent parameter corresponding to specular reflection. When leaf inclination angle is not considered, i.e., the leaf surface is perpendicular to the normal, the incidence angle (θ_s, the angle between the light source and the normal to the leaf) is equal to the illumination angle. In this case, PROCOSINE model can be simplified to the leaf DHRF simulated by PROSPECT-5 and specular parameter b

$$\mathrm{BRF}_{\mathrm{PROCOSINE}} = \mathrm{DHRF}_{\mathrm{PROSPECT-5}} + b. \tag{11}$$

The simulated BRF spectra of 764 leaves representing nadir direction ($b_0 = 0$) were generated initially, and the biochemical statistical characteristics are shown in Table III. Subsequently, four levels of specular reflections ($b_1 = 0$–0.15, $b_2 = 0.15$–0.3, $b_3 = 0.3$–0.45, $b_4 = 0.45$–0.6) were introduced to the BRF, which result in each sample corresponding to five simulated spectra.

These datasets, comprising 1919 samples for LCC and 2143 samples for EWT, encompass various measurement strategies (MA, integrating sphere, leaf clip measurements, and model simulation), plant species, and growth stages (with different leaf surface structures and extensive chlorophyll and water gradients). This diversity allows for a thorough verify the robustness and transferability of $\mathrm{SACI}_{\mathrm{BD}}$ proposed in this study.

D. Canopy Imaging Datasets Used for Validation

To further validate $\mathrm{SACI}_{\mathrm{BD}}$ models and compare them with $\mathrm{SACI}_{\mathrm{BRF}}$ and existing spectral indices at the canopy level, two canopy datasets were obtained using close-range camera and LESS simulated scene.

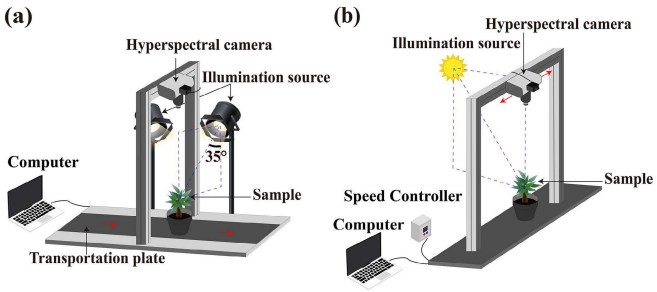


Fig. 2. Hyperspectral imaging setup in (a) laboratory and (b) field.

1) Close-Range Hyperspectral Images: A push broom Resonon Pika L Hyperspectral Camera (Spectronon Pro, Resonon, Bozeman, MT, USA) was utilized to acquire close-range hyperspectral images from plant canopies with different structures in the laboratory and field. The camera obtains 300-band images within the wavelength region of 375–1011 nm, and the sampling interval is about 2.1 nm. The distance from the camera and the canopy top was kept 0.5–1 m. The spatial resolution of the image is very high because the camera scans 900 pixels per line in a field of view of 13.1°. Two 1000 W halogen lamps covering the visible and near-infrared (VNIR) regions, were used for indoor measurements to provide smooth and monotonic spectrum stably [see Fig. 2(a)]. To calibrate the raw data in the image to the reflectance factors at each pixel, a Spectralon panel (white reference) was horizontally laid near the plant and captured for each image acquisition, following the conventional approach used in high-resolution imaging studies [39], [106]. The plant samples were placed on a low-reflectance (smaller than 0.05) black velvet fabric to minimize the influence of the background. The field measurements [see Fig. 2(b)] were carried out from 11:00 to 14:00 on July 20th and 25th, 2023, with the SZA changing from 25° to 36°. The sky was clear during the measurements. The reflection spectra extracted from the image data were used to evaluate the performance of $\mathrm{SACI}_{\mathrm{BD}}$ on LCC estimation, but the EWT-related analyses were unable to conduct due to the narrow wavelength range (375–1011 nm) of the hyperspectral sensor.

Twelve individual plants with different canopy structures were imaged, including eight plants [two plants of *Schefflera arboricola* (*Sa*), three plants of *Epipremnum aureum* (*Ea*), two plants of *Hydrocotyle vulgaris* L. (*Hv*), and one plant of *Peperomia magnoliifolia* (*Pm*)] in the laboratory and four plants (*Schefflera arboricola* (Sa), *Pachira aquatica* (*Pa*), *Epipremnum aureum* (*Ea*), and *Peperomia polybotrya*(*Pe*), one plant for each species) in the field. Sixty leaves (42 from the laboratory and 18 from the field) were cut from the canopy for LCC measurement immediately after the hyperspectral image acquisition. The $\mathrm{SACI}_{\mathrm{BD}}$ derived from the mean reflection spectra of each leaf was used to relate to LCC.

2) LESS Simulation Scene Image: To further validate $\mathrm{SACI}_{\mathrm{BD}}$ at the canopy level, a 3-D Radiative Transfer Model Intercomparison (RAMI) scene, HET14_WCO_UND, was introduced and simulated by the LESS model [63], [107].

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TABLE IV
BIOCHEMICAL PARAMETER COMBINATIONS (PROSPECT-5) AND SPECULAR REFECTION VALUES ASSIGNED TO EACH INDIVIDUAL TREE IN THE HET14_WCO_UND SCENE, AS SIMULATED BY LESS

Abbreviations	N	LCC (μg/cm^2)	Car (μg/cm^2)	EWT (g/cm^2)	LMA (μg/cm^2)	Specular reflection
CISI1	1.37	30.40	9.94	0.0196	0.0047	0.268
CISI2	0.91	26.14	6.56	0.0088	0.0057	0.295
CISI3	1.72	17.16	4.96	0.0128	0.0043	0.094
CISI4	1.67	47.47	8.72	0.0069	0.0045	0.257
CISI5	1.80	59.60	15.58	0.0363	0.0122	0.255
CISI6	1.14	15.90	4.10	0.0071	0.0075	0.231
CISI7	2.05	41.74	10.83	0.0251	0.0094	0.099
CISI8	2.03	32.96	9.82	0.0200	0.0085	0.010
CISI9	1.35	64.72	14.96	0.0107	0.0065	0.151
CISI10	1.29	39.53	6.19	0.0058	0.0030	0.002

This scene represents a 108.25 × 103.90 m Wellington Citrus Orchard located at 33°36' S, 18°56' E, comprising of ten individual Citrus trees arranged with an inter-row distance of 4.5 m [108]. To effectively validate the SAC_{BD} method, two customizations were applied to the scene: 1) given the original scene only included leaf optical properties and lacked the corresponding biochemical traits for each tree, the leaf optical properties of individual Citrus trees were assigned using PROSPECT-simulated reflection with known biochemical parameters (see Table IV) and 2) in high spatial resolution imagery, specular reflection was inevitable. Therefore, a random specular reflection constant (a uniform distribution between 0 and 0.3 shown in Table IV) was added to each PROSPECT-simulated reflection to represent the optical properties of Citrus tree leaves [43], [44]. The biochemical parameters of the PROSPECT were generated using a multivariate normal distribution [92]. The optical properties of branches, stems, and twigs were set to "birch_branch" from the LESS database, and the background was simulated using the GSV model with default settings (brightness parameter C1:0.3; shape parameters C2 and C3:0; water parameter CSM: −0.05) in LESS [109]. To validate SAC_{BD} on the HET14_WCO_UND scene, the simulation was conducted on a 20 × 20 m subplot (spatial resolution of 2.5 cm) using an orthographic imaging configuration across the 400- to 2500-nm spectral range [see Fig. 3(a)]. A total of 34 sample points from nine individual Citrus trees were selected following two steps [see Fig. 3(a)]. First, the center point of each tree within the scene was identified using the spatial positioning file (instances.txt) from the LESS. Then, to exclude potential soil or nonleaf samples, a spectral filtering criterion was applied: only points where the BRF at 670 nm was lower than at 560 nm were retained. The BRF spectra of these sample points [see Fig. 3(b)], which accounted for the influence of specular reflection and accompanied by biochemical parameters (LCC and EWT), were extracted to evaluate the performance of $SACI_{BD}$.

III. RESULTS

A. Spectral Properties of Leaf BRF and MA BD

The spectral characteristics of leaves from three species with distinct surface structural properties (*Ju*, *Pl*, and *Sm*) are shown in Fig. 4. For BRF [see Fig. 4(a)–(c)], wavelength-independent specular reflection produces additive difference in the anisotropic spectral without altering absorption features. Studies have confirmed that the BRF with maximum spectral intensity is usually detected near the forward specular direction [26]. The plant species with different leaf surface structures generate various magnitudes of specular reflections, thus affecting BRF.

The spectral characteristics of MA BD [see Fig. 4(d)–(f)] reveal the overall absorption capacity of leaves. The spectral shape of absorption peaks near 675 and 1950 nm, which are sensitive to the changes of chlorophyll and water in leaves, serves as the basis for estimating LCC and EWT using the SAC method. MA BD shows multiplicative characteristics in the absorption wavelengths (5), with the nonabsorption wavelengths normalized to equal values across different VZAs [see Fig. 4(d)–(f)]. The SAC method will eliminate such multiplicative characteristics caused by specular reflection in various VZAs and plant species. In general, leaf MA BD, with its absorption features and multiplicative spectral intensity differences, aligns well with the SAC method and is expected to support the high-precision estimation of LCC and EWT across different incident-observation geometries.

B. SACIBRF and $SACI_{BD}$ Based on BRF

The R^2 contour maps for LCC and EWT, based on the combination of SAC with MA BD ($SACI_{BD}$) and the combination of SAC with BRF ($SACI_{BRF}$), are shown in Fig. 5. The most effective OSIs (with the maximum R^2) for MA BD located at the absorption peaks of chlorophyll or water [OSI_{LCC}: 658–748 nm or OSI_{EWT}: 1872–2026 nm, white star in Fig. 5(b) and (d)], where the spectral shapes are sensitive to the changes of biochemical contents (see Fig. 6). Some SIs out of the absorption wavelengths also showed relatively high R^2, such as 1975–2035 nm [white triangle in Fig. 5(d)] with $R^2 = 0.89$ for estimating EWT. The maximum R^2 values for LCC and EWT obtained using MA BD [$SACI_{BD}$, Fig. 5(b) and (d)] are significantly higher than those achieved using BRF [$SACI_{BRF}$, Fig. 5(a) and (c)]. Moreover, the relationships between MA $SACI_{BRF}$ and LCC or EWT are sensitive to VZAs and show low R^2 in the

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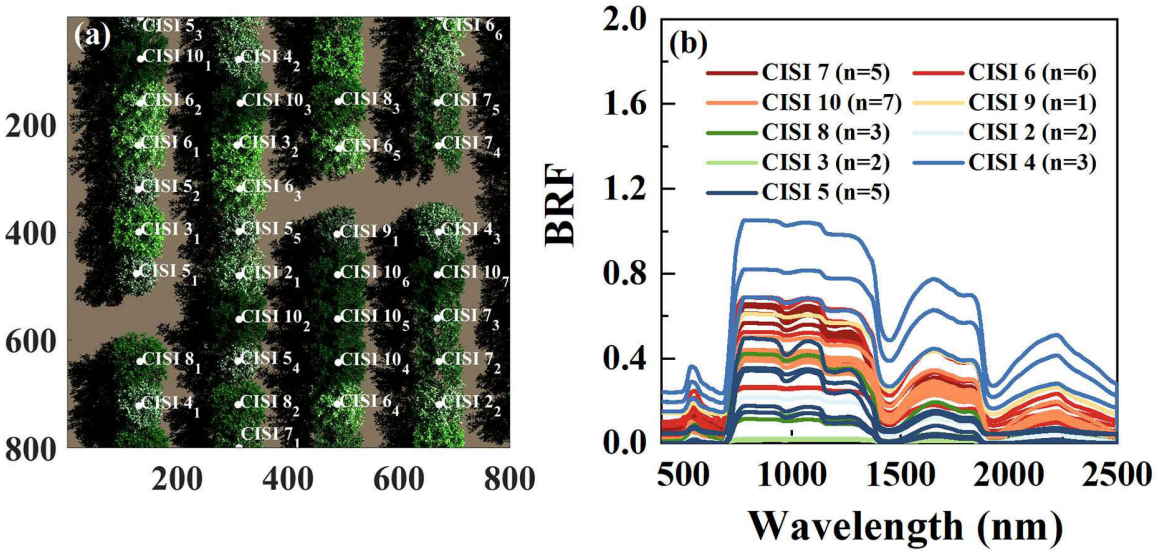


Fig. 3. 3-D RAMI scene simulated by LESS (HET14_WCO_UND). (a) Location of the 34 sample points extracted from nine citrus tree (labeled as "CISI a_b," where "CISI a" denotes the individual citrus tree, and subscript "b" represents the specific sample number for each citrus tree) and (b) corresponding BRF spectral for each sample point.

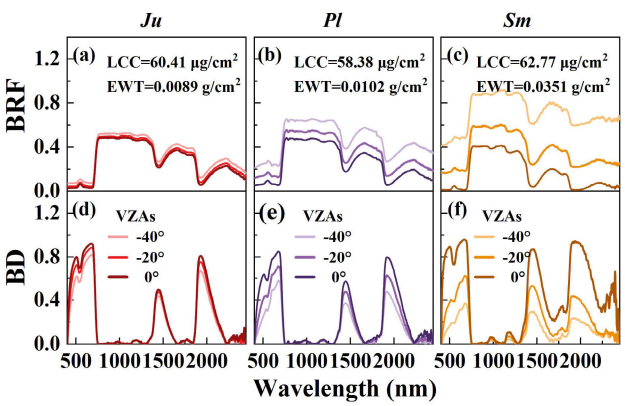


Fig. 4. (a)–(c) BRF and (d)–(f) MA BD of leaf samples from species with different surface properties (*Ju*, *Pl*, and *Sm*). Color scales correspond to spectra at 0°, −20°, and −40° VZAs ("−" represents the forward scattering directions).

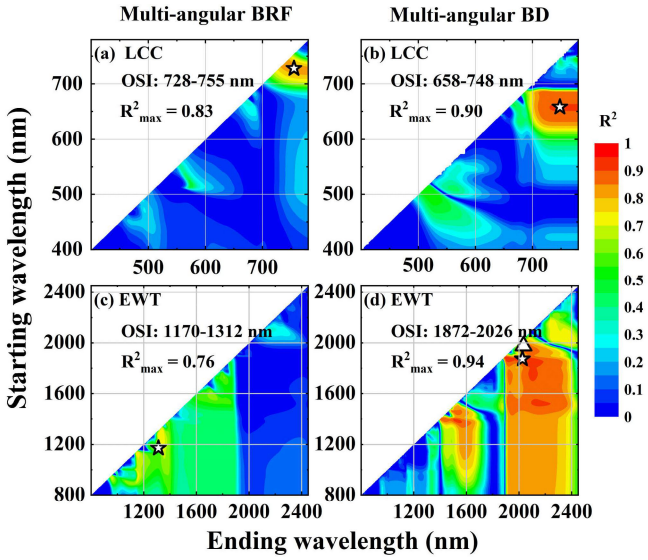


Fig. 5. R^2 contour maps of (a) and (b) LCC and (c) and (d) EWT using SAC method based on BRF [$SACI_{BRF}$, (a) and (c)] and MA BD [$SACI_{BD}$, (b) and (d)] from calibration dataset (labeled as "calibration dataset" in Table II). "Stars" indicate the locations of the OSI corresponding to the maximum R^2. White triangle (d) represents the location (1972–2035 nm) that showed relatively high R^2 (R^2 = 0.89).

forward scattering directions dominated by specular reflection [see Fig. 7(a) and (c)]. Therefore, the combination of SAC and BRF results in relatively weak relationships with LCC or EWT over all VZAs. In contrast, the similar high R^2 and fitting slopes of 13 VZAs [see Fig. 7(b) and (d)], as well as the stable R^2 across 13 VZAs under three SZAs [see Fig. 8(a) and (b)], illustrate that the combination of SAC and BD is not only strongly related to biochemical parameters but also insensitive to SZAs and VZAs. These results suggest the superiority of $SACI_{BD}$ in establishing the relationships with LCC and EWT; that is, they are not affected by the specular reflection caused by different leaf surface structural properties and variable incident-observation geometries. Therefore, $SACI_{BD}$ within OSIs and derived empirical models in Fig. 5(b) and (d) can be further applied to estimate LCC and EWT.

C. Validation of $SACI_{BD}$ and Existing Spectral Indices Based on Leaf Validation Datasets

Five leaf datasets (validation dataset in Tables II and III), including abundant plant species and different spectral measurement strategies, were applied to validate the generality and predictability of the $SACI_{BD}$-derived empirical models for LCC and EWT based on MA BD. The $SACI_{BD}$ showed high estimation accuracy on individual datasets and combined datasets (LCC: RMSE = 5.98 μg/cm^2; EWT: RMSE = 0.0031 g/cm^2) [see Fig. 9(a) and (b)], although the biochemical parameters of large values are slightly underestimated. For the

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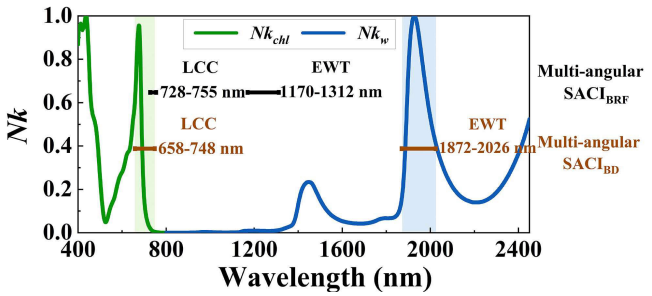


Fig. 6. Normalized specific absorption coefficient of chlorophyll or water (Nk_{chl} or Nk_w). The black bars indicate the OSI for MA SACI$_{BRF}$, while brown horizontal bars and shadow area represent the OSI for MA SACI$_{BD}$.

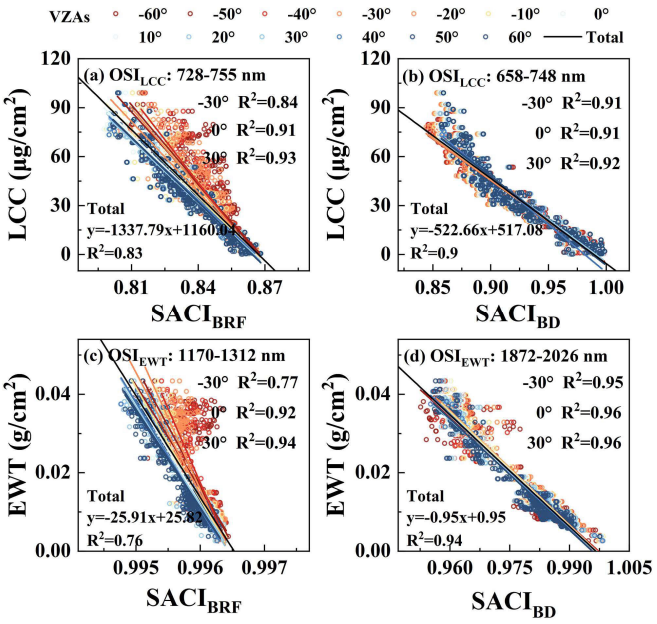


Fig. 7. Relationships between SACI$_{BRF}$ (the SAC combined with BRF) and (a) LCC or (c) EWT, and between SACI$_{BD}$ (the SAC combined with MA BD) and (b) LCC or (d) EWT at 13 different VZAs and for all VZAs (represented by "Total") based on calibration dataset (calibration dataset of Table II). R^2 and regression formulas from three representative VZAs (−30°, 0°, 30°, and "−" represents the forward scattering directions) or the summary of all VZAs are shown, respectively.

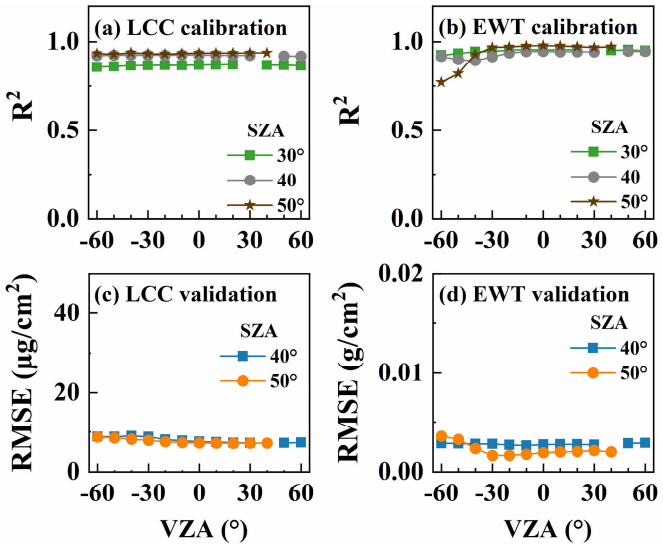


Fig. 8. Relationships of SACI$_{BD}$ and LCC or EWT for calibration samples [(a) and (b) calibration dataset of Table II] and validation samples [(c) and (d) sample number: validation dataset of Table II] at different SZA and VZAs in the principal plane.

MA validation dataset, the estimation accuracy is not only high (RMSE = 7.91 μg/cm^2 for LCC or 0.0027 g/cm^2 for EWT) but also insensitive to SZAs and VZAs, as shown in Fig. 8(c) and (d). These results confirm that the newly proposed SACI$_{BD}$ based on MA BD can established robust relationships with leaf biochemical parameters, enabling precise estimation of LCC and EWT across a wide range of plant species under various growth statuses and detection conditions.

The linear relationships between existing spectral indices and LCC or EWT, based on the MA reflectance factors of leaf samples over all the VZAs (n = 297 for LCC and n = 317 for EWT shown in Table II, labeled as "calibration dataset") and the nadir direction, are shown in Table V. For LCC estimation, SACI$_{BD}$ ranks second with an R^2 of 0.90, following the $(R_{720} - R_{761})/(R_{720} - R_{672})$, which exhibits the strongest correlation to LCC (R^2 = 0.93). Other difference ratio indices, such as $[(R_{850} - R_{710})/(R_{850} - R_{680})$, $(R_{750} - R_{445})/(R_{705} - R_{445})$, and $(R_{720} - R_{761})/(R_{720} - R_{672})]$, and derivative index ($D_{730}$), also showed strong relationships with LCC ($R^2 \geq 0.83$). This is due to their ability to mitigate some effects of the specular reflection, as demonstrated by the similar results obtained from both nadir direction (including weak specular reflection) and MA data. In contrast, the other types of spectral indices are not well related to LCC and show a significant decrease in R^2 when applied to MA data compared to nadir measurements, because most of them cannot remove the specular effect and are sensitive to the measurement directions. Similar to LCC estimation, most of the existing spectral indices show relatively weak relationships with EWT except for D or DR spectral indices, which exhibit stronger correlations with EWT (see Table V). Interestingly, SACI$_{BD}$ exhibits the strongest relationship with EWT with a high R^2 of 0.94. These results demonstrated that SACI$_{BD}$ can provide accurate LCC and EWT estimation when the spectral data are measured from various directions.

D. Performance of SACI$_{BD}$ in Estimating LCC and EWT in Canopy Hyperspectral Images

The effectiveness of the proposed SACI$_{BD}$ in estimating LCC and EWT was further tested using the reflection spectra derived from hyperspectral images at the canopy level (from close-range camera and LESS simulated scene). The results were also compared with those obtained from existing spectral indices (see Figs. 10 and 11). Scatter plots of estimated versus measured LCC for all the spectral indices are shown in Fig. 10. SACI$_{BD}$ achieved the highest accuracy in LCC estimation (RMSE = 6.98 μg/cm^2) [see Fig. 10(p)]. The estimated values using SACI$_{BD}$ were closely distributed around the 1:1 line [see Fig. 10(p)], while most of the other spectral indices (except for $(R_{750} - R_{445})/(R_{705} - R_{445})$: RMSE = 7.28 μg/cm^2 and $(R_{750} - R_{705})/(R_{750} + R_{705} - 2^*R_{445})$: RMSE = 6.98 μg/cm^2) were more deviated from the 1:1 line [see Fig. 10(a)–(o)]. For EWT estimation, SACI$_{BD}$ also achieved the highest accuracy,

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TABLE V

COEFFICIENT OF DETERMINATION (R^2) AND LINEAR REGRESSION EQUATIONS TO SHOW THE RELATIONSHIPS BETWEEN SPECTRAL INDICES AND LCC OR EWT BASED ON THE MA DATA OVER ALL THE VZAS AND IN THE NADIR DIRECTION. ALL REGRESSIONS ARE STATISTICALLY SIGNIFICANT ($P < 0.001$). R_Λ REPRESENTS BRF AT THE WAVELENGTH Λ

	LCC				EWT			
Type	**Index**	**R^2**	**Regression equation**	**R^2 at 0°**	**Index**	**R^2**	**Regression equation**	**R^2 at 0°**
SR	R_{750}/R_{550}	0.26	y=5.86x+15.13	0.53	R_{1300}/R_{1450}	0.44	y=0.003x+0.007	0.78
	R_{860}/R_{550}	0.27	y=5.79x+15.29	0.54	R_{900}/R_{970}	0.51	y=0.387x-0.38	0.73
	R_{750}/R_{710}	0.68	y=28.25x-18.32	0.89	R_{860}/R_{1240}	0.21	y=0.085x-0.075	0.46
					R_{1670}/R_{1880}	0.4	y=0.006x+0.004	0.76
					R_{860}/R_{1650}	0.46	y=0.026x-0.02	0.85
ND	$(R_{750}-R_{705})/(R_{750}+R_{705})$	0.65	y=101.4x-0.72	0.86	$(R_{860}-R_{1240})/(R_{860}+R_{1240})$	0.19	y=0.175x+0.011	0.44
	$(R_{800}-R_{650})/(R_{800}+R_{650})$	0.23	y=48.43x+4.26	0.51	$(R_{820}-R_{1600})/(R_{820}+R_{1600})$	0.44	y=0.074x+0.002	0.86
	$(R_{800}-R_{635})/(R_{800}+R_{635})$	0.31	y=55.78x+1.53	0.60	$(R_{860}-R_{1640})/(R_{860}+R_{1640})$	0.41	y=0.078x+0.003	0.84
					$(R_{1062}-R_{1393})/(R_{1062}+R_{1393})$	0.59	y=0.001x+0.066	0.94
D or DR	D_{730}	0.88	y=7358.9x-3.12	0.90	$R_{1065}-R_{1310}$	0.64	y=0.395x	0.86
	$(R_{850}-R_{710})/(R_{850}-R_{680})$	0.83	y=101.63x-15.25	0.85	$(R_{850}-R_{2218})/(R_{850}-R_{1928})$	0.47	y=0.058x-0.022	0.80
	$(R_{750}-R_{445})/(R_{705}-R_{445})$	0.89	y=13.49x-7.89	0.92	$(R_{850}-R_{1788})/(R_{850}-R_{1928})$	0.38	y=0.041x-0.001	0.86
	$(R_{720}-R_{761})/(R_{720}-R_{672})$	0.93	y= -65.46x-1.1	0.94	$(R_{1271}-R_{410})/(R_{1342}-R_{410})$	0.91	y=0.132x-0.134	0.96
Others	$(R_{750}-R_{705})/(R_{750}+R_{705}-2*R_{445})$	0.86	y=103.52x-11.73	0.87	$2*R_{1530}-R_{1005}-R_{2055}$	0.76	y=-0.125x+0.005	0.92
	$R_{705}/(R_{717}+R_{491})$	0.74	y=-180.89x+135.8	0.80	$2*R_{1405}-R_{925}-R_{1885}$	0.76	y=-0.111x+0.001	0.88
	$1/R_{700}$	0.38	y=3.21x+14.14	0.64				
	$1/R_{550}-1/R_{750}$	0.21	y=2.37x+23.31	0.44				
	$1/R_{700}-1/R_{750}$	0.42	y= 3.51x+19.16	0.67				
$SACI_{BD}$	**LCC-OSI: 658-748 nm**	**0.9**	**y=-522.66x+517.08**	**0.91**	**EWT-OSI: 1872-2026 nm**	**0.94**	**y=-0.952x+0.954**	**0.96**

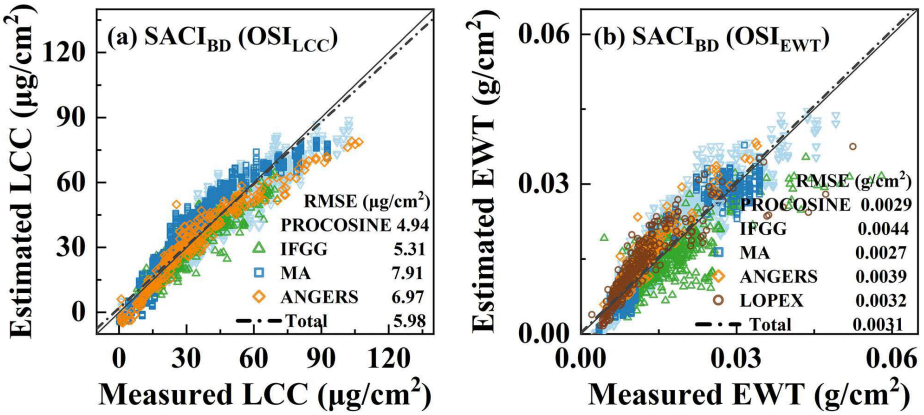


Fig. 9. LCC or EWT estimated by $SACI_{BD}$ derived empirical models versus measured values of five validation datasets in Tables II and III (MA, LOPEX, ANGERS, IFGG, and PROCOSINE). (a) and (b) RMSE (μg/cm^2 for LCC or g/cm^2 for EWT) are shown, and "Total" represents the combination of all datasets. For the MA validation dataset, the RMSE of LCC or EWT at 13 VZAs are displayed in Fig. 8(c) and (d).

with RMSE of 0.0044 g/cm^2 [see Fig. 11(p)]. These results confirm the effectiveness of $SACI_{BD}$ for estimating LCC and EWT on the canopy level.

Moreover, the $SACI_{BD}$ can be applied to estimate LCC for each pixel in hyperspectral image (smoothed using Savitzky–Golay method, second-order and 20-nm bandwidth window). The resulting mapped LCC distribution for the close-range image, based on relationships derived from hyperspectral imaging data captured with a close-range camera ($n = 60$) is shown in Fig. 12. Due to the combination of the leaf inclination and viewing directions, which results in the specular reflection dominates a part of leaf surface, some pixels appeared very white and seemed glittering on the true color composite image (450, 550, and 680 nm) in Fig. 12(a) (e.g., leaf 1 and leaf 2). To make clear the capability of $SACI_{BD}$ in reducing the specular effect, the areas with large parts of white pixels and the corresponding estimated LCC by some selected spectral indices were circled in red in Fig. 12. It is easy to see that the LCC was evenly estimated and the LCC values do not change as sharply on the hyperspectral image [see Fig. 12(e)]. The mean LCC estimated using $SACI_{BD}$ closely aligns with the measured LCC [as shown in the distribution in Fig. 12(e)]. However, the estimated LCC values were noticeably affected by the intensity of reflection in the image when using other spectral indices, which caused significant overestimation or underestimation in the specular region of interest (ROI), and a substantial deviation between the mean estimated LCC and the measured LCC [see Fig. 12(b) and (d)]. To quantify the influence of specular reflection and shadow on LCC estimation, a specular and a shadow ROI are outlined in white [see Fig. 12(a)], and their corresponding LCC estimates are labeled around the white circle [see Fig. 12(b)–(e)]. The BRF in the specular reflection ROI increased compared to that of leaf 1, while the BRF in the shadow ROI decreased

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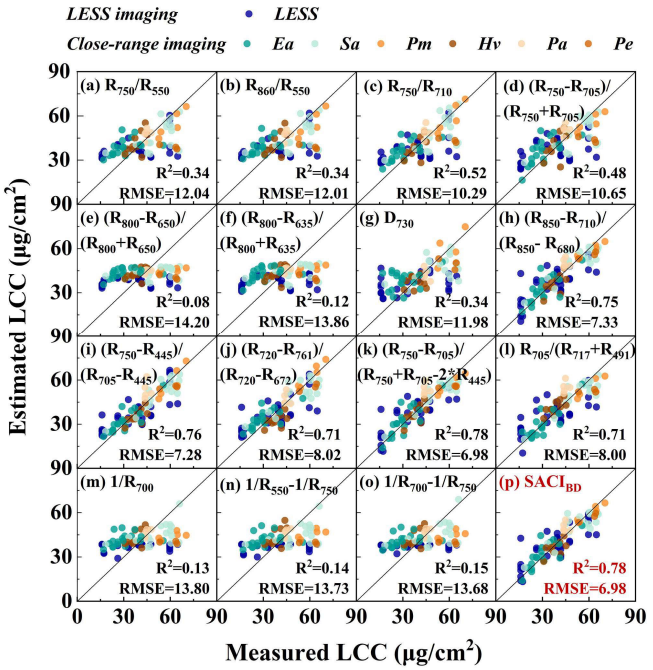


Fig. 10. Scatter plots of measured and estimated LCC (μg/cm^2) with different spectral indices using hyperspectral imaging data from close-range camera (n = 60) and LESS simulation scene (HET14_WCO_UND, n = 34). (a)–(o) Correspond to existing spectral indices (Table I), categorized as (a)–(c) SR, (d)–(f) ND, (g)–(j) D or ND, and (k)–(o) other types. (p) Represents the SACIBD.

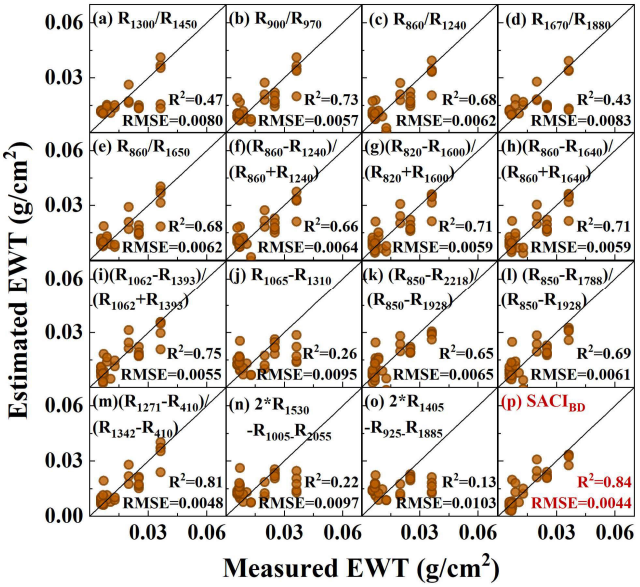


Fig. 11. Scatter plots of measured and estimated EWT (μg/cm^2) with different spectral indices using hyperspectral imaging data from LESS simulation scene (HET14_WCO_UND, n = 34). (a)–(o) Correspond to the existing spectral indices (Table I), categorized as (a)–(e) SR, (f) and (i) ND, (j)–(m) D or ND, and (n) and (o) other types. (p) Represents the SACIBD.

compared to that of leaf 3. These BRF variations caused by specular reflection and shadow may affect LCC estimation accuracy. D_{730} showed unsatisfactory inversion results across the image [see Fig. 12(b)], including the specular and shadow ROIs. The $1/R_{700} - 1/R_{750}$ were significantly affected by specular reflection, with estimated LCC deviations of 7 and 14 μg/cm^2 from its estimated mean LCC and measured mean LCC, respectively [see Fig. 12(d)]. $(R_{750} - R_{445})/(R_{705} - R_{445})$ has been considered to be able to reduce some specular reflection, and exhibited slight sensitivity to specular reflection but was considerably influenced by shadow-induced noise, resulting in a 10 μg/cm^2 deviation from its estimated mean LCC (Fig. 12(c). Meanwhile, the LCC mapping by $(R_{750} - R_{445})/(R_{705} - R_{445})$ was not as even as that of $SACI_{BD}$, which indicated that $SACI_{BD}$ may be more insensitive to specular influence. In contrast, $SACI_{BD}$ remained robust against both specular and shadow effects, with an estimated LCC deviation of less than 3 μg/cm^2 [see Fig. 12(e)].

The mapped LCC and EWT distributions based on the LESS simulation scene are presented in Fig. 13. $SACI_{BD}$ effectively retrieved LCC and EWT despite the presence of specular reflection, showing clear spatial variations among samples. The inversion results exhibit a gradual transition in color from high to low LCC values [see Fig. 13(a)], as observed in CISI 9 (64.72 μg/cm^2), CISI 5 (59.60 μg/cm^2), CISI 10 (39.53 μg/cm^2), and CISI 6 (15.90 μg/cm^2). A similar trend is found in the EWT inversion map [see Fig. 13(b)], with values decreasing from CISI 5 (0.0363 g/cm^2) to CISI 7 (0.0251 g/cm^2), CISI 3 (0.0128 g/cm^2), and CISI 6 (0.0071 g/cm^2). These results demonstrate the potential of $SACI_{BD}$ as a practicable tool to map leaf biochemical parameters at canopy level.

IV. Discussion

MA measurement is an essential aspect of the study of leaf spectroscopy [39], [49], [99]: on the one hand, the directional reflectance factors of leaves are objectively anisotropic in hemispherical space, which is most evident in the forward specular directions [26], [27], [29]; on the other hand, remote sensing detections, such as the imaging methods with high spatial resolution, inevitably contain reflection information from different directions, because the pixel is specific to a small part of an individual leaf with random inclinations under arbitrary solar incident geometries [59], [110]. Such different reflections of leaves varying with observation directions are mainly caused by wavelength-independent specular reflections [32], [38]. Since there is no interaction with internal biochemistries, the specular components generated from the leaf surface do not carry biochemical information and will interfere with the estimation of leaf biochemical parameters [16], [60]. Based on MA hyperspectral measurements, this study paid attention to and finally overcame the specular reflection influence on estimating LCC and EWT using the combination of SAC and BD. The $SACI_{BD}$ method is derived from leaf level and validated on images with a spatial resolution finer than the leaf size, where pixels can be considered pure. To minimize potential uncertainties introduced by mixed pixels, we recommend applying $SACI_{BD}$ method to images with a resolution finer than the leaf size.

SAC, a spectral shape-based metric, was used to quantify the spectral shape similarity between MA BD and Nk_{chl} or Nk_w in the OSIs, without being affected by specular reflection. On this basis, the newly developed $SACI_{BD}$ successfully predicted LCC and EWT across various VZAs. MA BD shows multiplicative effects among VZAs (5), which can be further proved by the approximate constant of the ratio of BD in a specific VZA to that in nadir (see Fig. 14). In this case, the principle of the SAC method was well applied [51], [54],

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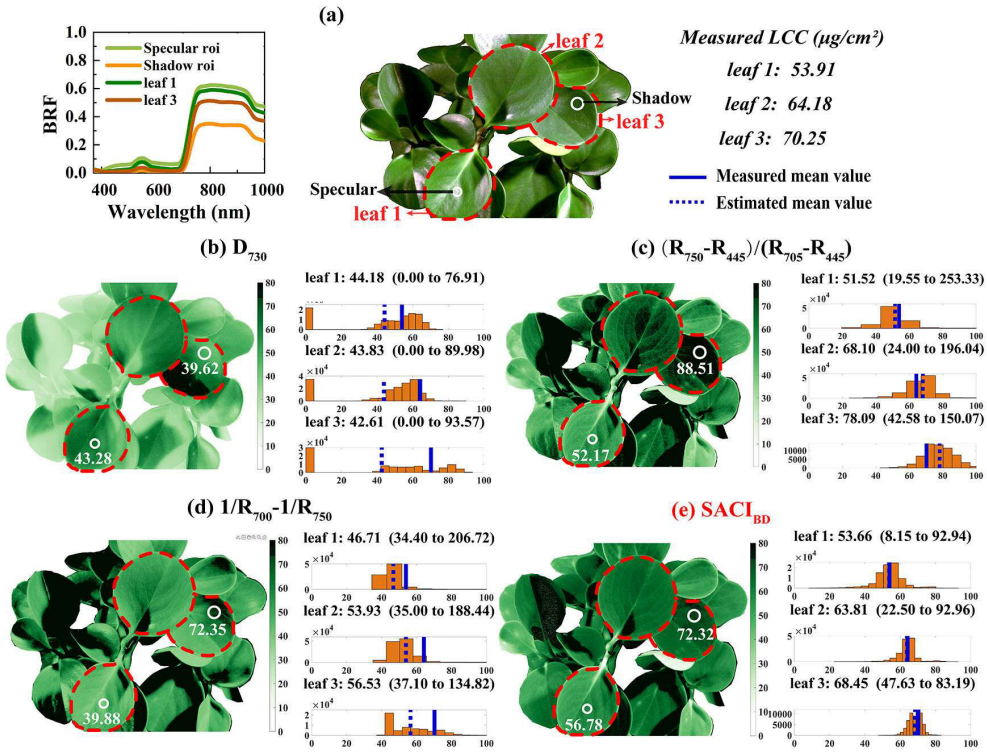


Fig. 12. (a) True color composite image, as well as estimated LCC and the estimation LCC distribution of three leaves (b) D_{730}, (c) $(R_{750} - R_{445})/(R_{705} - R_{445})$, (d) $1/R_{700} - 1/R_{750}$, and (e) $SACI_{BD}$. The sample is *Pe*. Measured LCC of leaf 1, leaf 2, and leaf 3 was 53.91, 64.18, and 70.25 $\mu g/cm^2$, respectively [as shown in image (a)]. The labels in the distribution represent the mean value (minimum and maximum). The BRF of specular and a shadow ROI are given in image (a), with the corresponding LCC estimates are labeled around the white circle in (b)–(e). Leaves (leaf 1 and leaf 2) with specular reflection (white pixels) obviously decrease the LCC estimation for (b) D_{730} and (d) $1/R_{700} - 1/R_{750}$. The estimation accuracy of (c) $(R_{750} - R_{445})/(R_{705} - R_{445})$ is slightly smaller than $SACI_{BD}$ in canopy image in this study. $SACI_{BD}$ has the best estimation of LCC.

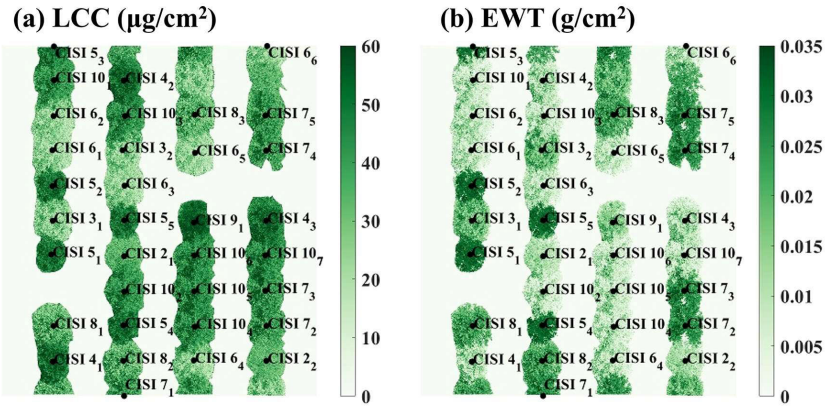


Fig. 13. Estimated (a) LCC and (b) EWT using $SACI_{BD}$ based on hyperspectral imaging data from LESS simulation scene (HET14_WCO_UND), based on model calibrated by 34 sample point (Figs. 10 and 11).

[57], that is, the cosα between target and reference spectral vectors does not change with the multiplicative gain of target spectrum [see Fig. 1(e) and (f)]. Thus, $SACI_{BD}$ was insensitive to VZAs. In addition, the OSIs stably located at the chlorophyll and water absorption peaks of BD (as "target spectra"), which become higher and flatter with increased LCC or EWT (see Fig. 15) [111], [112]. This may be because the absorption becomes saturated when the biochemical contents are high, according to the power exponential relationships between the absorption and biochemistry concentration in Lambert–Beer Law. Generally, the saturation starts from the absorption peak and spreads to both sides of the peak, which results in the broader absorption bulge plateau and dissimilarity with the spectral shape of *Nk* (as "reference spectrum"), and when leaf biochemical contents reach a certain level, the change of spectral shape caused by absorption will be less significant due to saturation. In fact, *Nk* represents a single molecule's absorption and can only be obtained when the chlorophyll or water concentration is low, so the BD with high LCC and EWT should have less similarity with *Nk*. Based on the principle of the SAC method, the less the spectral shape similarity between target and reference, the greater the α, and the smaller

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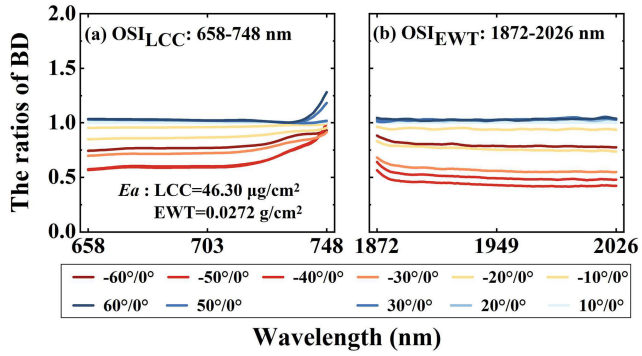


Fig. 14. Ratios of BD in different VZAs to that in nadir, which prove the multiplicative characteristics of MA BD in (a) OSI$_{LCC}$ (658–748 nm) and (b) OSI$_{EWT}$ (1872–2026 nm), taking an *Ea* leaf sample (40° SZA) as an example.

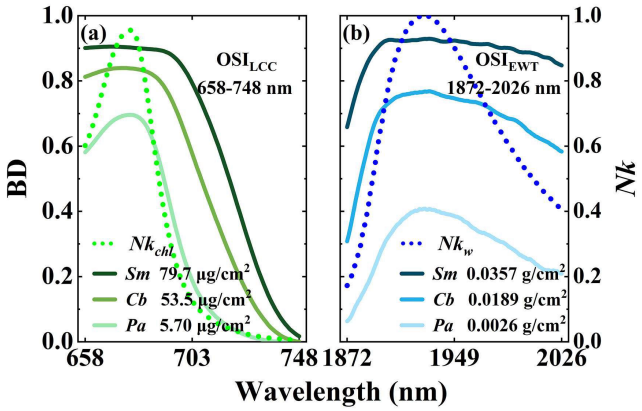


Fig. 15. Details of BD with (a) LCC gradients and Nk_{chl} in OSI$_{LCC}$ (658–748 nm) and (b) EWT gradients and Nk_w in OSI$_{EWT}$ (1872–2026 nm). Six leaf samples from *Sm*, *Cb*, and *Pa* are displayed.

the cosα between their spectral vectors; thus, the significant negative relationships between SACI$_{BD}$ and LCC or EWT can be understood. From this point of view, it seems that the BRF and SAC method combination is not entirely reasonable, although good estimation accuracy of EWT was obtained in a previous study [58]. The absorption and reflection must have a negative relationship explained by Fang et al. [58], but there may be a lack of theoretical consideration of the changes in absorption shapes of samples with different biochemical contents. On the one hand, the spectral shape of reflection does not match the absorption, i.e., *Nk*, because they are inversely corresponding to each other. On the other hand, the additive differences in the spectral intensity among VZAs cannot be eliminated entirely by the SAC method, such as in the example in Fig. 1(c) and (d). In general, SAC combined with BD in the absorption wavelengths showed great potential in estimating LCC and EWT from varied VZAs.

The simple linear models calibrated from leaf MA measurement dataset, which encompasses diverse plant species, biochemical gradients, and multiple incident-observation geometries, showed robustness and universality. Thus, these models can be applied to the simulated or measured leaf datasets from various, regardless of measurement strategy, regions, and environments. These excellent results will support rapid and nondestructive biochemical monitoring of botany or ecology based on hyperspectral remote sensing technology. Although OSI$_{EWT}$ (1872–2026 nm) may not be effective in the field detection scenarios due to the absorption of water vapor in the atmosphere [71], [113], the other region [OSI$_{EWT}$: 1975–2035 nm, $R^2 = 0.89$; Fig. 5(d)], which is outside of the absorption of atmosphere vapor, also shows advantages compared with previous studies that used spectral reflection information and spectral indices to relate with EWT [71], [93], [114]. Alternative OSI$_{EWT}$ region enables SACI$_{BD}$ to be suitable for different measurement conditions, such as laboratory and field. SACI$_{BD}$ is suitable for the spectral information taken from MA, integrating sphere, leaf clip measurements, or model simulation [35], [36], [93]. Moreover, SACI$_{BD}$ has been scaled up to estimate LCC and EWT at canopy level based on imaging datasets obtained through close-range camera and LESS simulated scene (see Figs. 10–13). Unlike traditional spectral indices, which utilize only a few bands, SACI$_{BD}$ makes full use of the continuity and high resolution of hyperspectral data to realize the high-precision estimation of both chlorophyll and water in leaves. These validations of SACI$_{BD}$ at both leaf and canopy scales demonstrated its effectiveness across various detection scales, indicating its potential for broader applications in estimating vegetation parameters from hyperspectral data at local to regional scales.

V. Conclusion

In this article, the combination of SAC and BD (SACI$_{BD}$) was proposed to realize high-precision estimation of leaf chlorophyll and water contents under different incident-observation geometries by mitigating the influence of specular reflection from the leaf surface. The superiority of combining SAC and BD was explained from two theoretical aspects: MA BD transformed additive specular differences into multiplicative characteristics and effectively extracted biochemical-relevant absorption features; SAC method utilized these chlorophyll and water absorption features in MA BD to accurately estimate LCC and EWT. Based on laboratory MA hyperspectral measurements, SACI$_{BD}$ within the biochemical-meaningful OSI$_{LCC}$ (658–748 nm) and OSI$_{EWT}$ (1872–2026 nm) were proved to have significant relationships with LCC or EWT without being affected by VZAs in the principal plane. In addition, the robustness and universality of the simple derived models enabled SACI$_{BD}$ to estimate LCC and EWT with high accuracy under various spectral measurements and plant species conditions. When the study was extended to vegetation canopy level, the good LCC estimation based on images acquired by close-range hyperspectral camera and LESS simulated scene further proved the effectiveness of the proposed SACI$_{BD}$. Therefore, the novel SACI$_{BD}$ will provide a convenient tool for efficient and nondestructive biochemistry monitoring of agricultural and forestry applications.

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